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A Stability-Centric Framework for Lightweight and Explainable Intrusion Detection

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Abstract

Effective intrusion detection for Internet of Things (IoT) environments requires balancing predictive performance, resource efficiency, and interpretability—particularly in real-world deployments where traffic distributions and attack scenarios vary. While many studies report near-perfect detection on benchmark datasets, this often overlooks model stability under distribution shifts. This paper addresses this gap by introducing a stability-focused evaluation of lightweight, explainable intrusion detection models using compact IoT-23 scenarios and a constrained set of 14 connection-level features for interpretability. Four lightweight models—logistic regression, random forest, XGBoost, and LightGBM—are assessed within a unified pipeline. Beyond standard internal validation, we implement a strict cross-scenario evaluation framework featuring a fully unseen malware capture. Our proposed Internal–External Stability Gap (IESG) framework, enhanced with normalized and multi-metric measures, highlights the degradation in consistency between internal and external metrics. Surprisingly, even models with high internal F1 scores (up to 0.9994) may experience considerable drops in external macro-F1 and specificity, exposing weaknesses in conventional evaluation. Experimentally, LightGBM provides the best trade-off between performance and compactness (606 KB) and shows the smallest stability gap for malicious detection. Nevertheless, all models show reduced balanced performance under scenario shift, underscoring that deployment readiness hinges on stability under changing conditions. Feature ablation reveals that leveraging high-impact features, such as port information, can boost internal accuracy at the expense of generalization. In summary, we demonstrate that while lightweight models deliver strong detection, only those proven stable across scenarios are viable for real-world IoT intrusion detection. Our evaluation framework offers a practical, interpretable tool for assessing model robustness.

Keywords: IoT intrusion detection; lightweight machine learning; explainable AI; IoT-23; cross-scenario validation; network security; security systems; Internet of Things (IoT); security

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1. Introduction

The motivation for this work stems from a practical deployment caveat: in Internet of Things (IoT) scenarios, Intrusion Detection Systems (IDSs) are frequently required to run at the gateway or near the network edge, where memory, disk storage, and processing resources are inherently constrained. Furthermore, IoT traffic differs from traditional

network datasets due to the vast number of low-cost devices generating it. Consequently, relying solely on signature-based monitoring fails to scale effectively to accommodate this diversity [1,2].

Public benchmarks allow reproducible testing of these challenges. For example, the CICIoT2023 dataset captures large-scale traffic from many devices and covers 33 attack types in seven classes [3]. The IoT-23 dataset is also relevant because it uses scenario-based captures from actual malware and real IoT devices, not just simulations [4].

Studies using the ToN_IoT dataset show that inconsistencies in feature definitions and attacks make it hard to compare results, even when performance appears high [5]. Still, similar methods persist in IoT IDS research. Most studies train and test several classifiers on a single dataset and report near-perfect accuracy. But they rarely check whether the features are concise enough for lightweight deployment, whether the models are clear to analysts, or whether the results hold up with new malware. Many surveys stress explainability, lightweight systems, and robustness, but do not assess these together [6–11].

This gap matters because an IDS can perform well on known data but fail under real-world conditions. One comparative study showed that accuracy drops as test sets become harder [12].

Similarly, research on attack-adaptive IoT network intrusion detection emphasizes that standard testing provides only a snapshot of a detector's utility, and that its true operational value depends on its resilience to shifts in network attacks [13]. In this study, we aim to bridge a narrowly defined yet highly practical gap in the literature. Rather than proposing a novel deep learning architecture, we investigate the extent to which detection performance is impacted by (i) intentionally restricting the feature set solely to connection-level attributes, (ii) limiting the model suite to lightweight classical and boosted-tree baselines, and (iii) conducting a final held-out evaluation on a traffic scenario absent from the training phase. Unlike prior work, this study explicitly incorporates feature stability and detection robustness as core components of the evaluation methodology.

The primary contributions of this work are summarized as follows:

- A lightweight, leakage-aware IoT intrusion detection framework based on 14 interpretable connection-level features, unified evaluation of LR, RF, XGBoost, and LightGBM under consistent validation, and an extended IESG framework quantifying robustness to distribution shifts.
- A unified evaluation of Logistic Regression (LR), Random Forest (RF), XGBoost, and LightGBM under identical preprocessing and cross-scenario validation conditions.
- An extended Internal–External Stability Gap (IESG) framework designed to quantify model robustness under distribution shifts.
- A SHAP-based explainability and feature-ablation analysis, focusing on port-related feature impact on model generalization. Deployment-oriented analyses balance predictive performance, compactness, efficiency, and robustness.
- A deployment-oriented analysis that balances predictive performance, model compactness, inference efficiency, and robustness.

It is important to note the defined scope of this investigation. This study does not aim to present state-of-the-art results on any specific IoT dataset, nor does it present high performance on a single train–test split as sufficient justification for real-world deployment readiness. Instead, it provides a lightweight framework that can be rapidly trained, comprehensively reviewed, and rigorously evaluated according to clearly defined stability and robustness metrics. By separating internal evaluation from stress testing on unknown data, we illustrate that realistic performance assessments for IoT IDSs must account for distribution shifts across varying malware scenarios. The remainder of this paper is structured as follows: Section 2 discusses literature Review and establishes the literature gap. Section 3

details the dataset selection and modeling process. Section 4 presents the results of the internal and external evaluations, alongside the feature ablation and stability analyses. Finally, Section 5 discusses these findings and their implications for lightweight intrusion detection in edge-oriented IoT systems.

2. Literature Review

Recent research on IoT intrusion detection can be broadly categorized into four main threads. The first thread focuses on assembling the benchmark dataset. For instance, the CICIoT2023 dataset expands both the volume of attacks and the diversity of represented devices [3]. Similarly, IoT-23 enhances the realism of malicious and benign traffic captures by linking them to specific execution scenarios [4]. Research using the ToN_IoT dataset highlights a closely related challenge: inconsistent feature policies and inconsistent attack labeling across datasets complicate cross-study comparisons, even when researchers have access to larger pools of traffic captures [5].

The second thread, in contrast, centers on model explainability. In production environments, alerts are difficult to trust unless analysts can inspect the underlying traffic characteristics driving them. Consequently, the feature sets utilized for IoT IDSs are increasingly required to meet this explainability constraint [7]. Surveys examining explainable AI for IoT and cyber analytics reach a consensus: post hoc explanations offer limited value unless they are carefully tied to semantic features and specific analytical questions, rather than being applied as an afterthought [2,9,14–16].

Moving to the third thread, the focus shifts to deployment efficiency. Numerous studies on lightweight IDSs, alongside boosting-based comparative analyses, demonstrate that carefully engineered models with small resource footprints can achieve highly competitive performance [10,11]. Furthermore, the development of hardware-aware pruning frameworks, ensemble tree methods, and Convolutional Neural Network (CNN) detectors tailored for compact Industrial IoT (IIoT) devices emphasizes that model size, inference latency, and feature extraction overhead are as critical to operational viability as predictive accuracy [17–20].

A fourth, closely related line of inquiry, builds upon deployment considerations and focuses on model behavior under distribution shifts. Analyses advocating for fair comparison conditions, methodologies evaluating attack-adaptive behavior, and recent surveys on federated intrusion detection all argue a similar point: dataset-wide evaluations relying on a single internal train-test split fail to adequately quantify predictive stability under distribution shifts [8,12,13,21,22]. While recent proposals have successfully increased accuracy through hybrid deep learning models, semantically enriched training data, and more efficient dimensionality reduction pipelines, they do little to address the diagnosis of performance degradation when the operational context changes [23–25].

A key concern in IoT intrusion detection research is the vulnerability of machine learning-based security systems to adversarial manipulation. IDS models are susceptible to attacks such as adversarial perturbations, poisoning, spoofing, traffic obfuscation, and protocol-level evasion, which can undermine detection in operational settings. These issues are particularly serious in cyber-physical environments, where attackers can disguise malicious traffic. The literature emphasizes that high predictive accuracy does not guarantee practical robustness. Researchers, therefore, call for robustness-aware evaluation methods that account for adversarial threats and distribution shifts [26].

Building on concerns about adversarial robustness, intrusion detection research is exploring decision fusion and ensemble coordination. Instead of one classifier, multiple models are combined using hard voting, soft voting, weighted fusion, or stacking to improve robustness and real-time stability. These methods show promise for industrial control

and cyber-physical environments, where balancing false positives and threat sensitivity is critical. Research also finds that fusion increases resilience under diverse traffic by reducing individual model bias and sensitivity to features [27].

Research Gap

A significant opportunity exists at the convergence of these four research areas. At present, there is a scarcity of research that simultaneously applies connection-level feature constraints, restricts baseline comparisons to lightweight models, includes explicit explainability, and tests models against external scenarios that are stricter than typical internal splits. To address this, this paper combines these elements into a concise, reproducible study using public IoT-23 data, carefully analyzing the performance outcomes. Therefore, the main contribution of this research lies in its methodological rigor and comprehensive evaluation, rather than in introducing new architectures. As shown in Table 1, there is a clear gap in the literature for unified frameworks that address evaluation standards, model interpretability, efficiency, and robustness simultaneously. Ultimately, this highlights the necessity for a more comprehensive and balanced methodology in designing Intrusion Detection Systems (IDS).

Table 1. Summary of Related IoT IDS Approaches.

Ref.	Scheme/Approach	Advantages	Limitations
[3]	CICIoT2023 Dataset	Large-scale dataset with diverse attacks and wide device coverage; suitable for comprehensive evaluation	Inconsistent feature definitions and labeling reduce cross-dataset comparability
[4]	IoT-23 Dataset	Realistic traffic with scenario-based labeling; high practical relevance	Limited scope and lack of standardized feature representation
[5]	ToN_IoT Framework	Highlights dataset heterogeneity and need for feature standardization	Does not fully solve interoperability across datasets
[7]	Explainable Feature Set	Improves transparency by linking decisions to meaningful features	Possible trade-off between interpretability and accuracy
[9,14,18,24]	Explainable AI (XAI) Approaches	Enhance trust through semantically meaningful explanations	Reliance on post hoc methods may not fully reflect model behavior
[10]	Lightweight IDS Models	Low computational cost; suitable for resource-constrained IoT environments	Reduced performance in complex or evolving attack scenarios
[11]	Boosting-Based IDS	High detection accuracy via ensemble learning	Increased computational complexity
[15]	Hardware-Aware IDS (HH-NIDS)	Optimized for deployment constraints and system efficiency	Dependence on specific hardware configurations
[16,17]	Ensemble Tree-Based Models	Strong accuracy and robustness	High complexity and reduced interpretability
[18,25]	Compact IIoT IDS	Low latency and small model size; suitable for real-time use	Possible loss of fine-grained detection capability
[8,12]	Robustness and Fair Evaluation	Emphasize fair benchmarking and performance validation	Limited generalization across unseen environments
[13]	Attack-Adaptive IDS	Adapts to evolving threats using incremental learning	Increased system complexity
[19,20]	Federated Learning IDS	Privacy-preserving and decentralized learning	Communication overhead and model inconsistency issues

Table 1. *Cont.*

Ref.	Scheme/Approach	Advantages	Limitations
[21]	Hybrid Deep Learning Models	High accuracy for binary and multi-class detection	High computational cost and low interpretability
[22]	Semantic/Self-Supervised Models	Improved feature representation; reduced need for labeled data	Complex training pipelines
[23]	PCA-Based Lightweight Ensemble	Efficient real-time detection with reduced feature space	Risk of information loss due to dimensionality reduction
[26]	Adversarial Robustness Review for IoT Smart Grids	Highlights vulnerabilities to adversarial perturbation, poisoning, and feature spoofing attacks	Primarily conceptual review without deployment-oriented evaluation framework
[27]	Decision Fusion IDS for Industrial Control Systems	Improves robustness and real-time stability using model-fusion strategies	Increased coordination complexity and higher computational overhead

3. Methodology and Materials

This section outlines the research methodology, including data preparation, feature and model selection, and validation for both internal and cross-scenario evaluations. Figure 1 displays the sequential workflow of the proposed lightweight, explainable IoT intrusion detection system: leakage-aware preprocessing is performed first, followed by compact feature extraction, model training, cross-scenario evaluation, stability analysis, and deployment considerations.

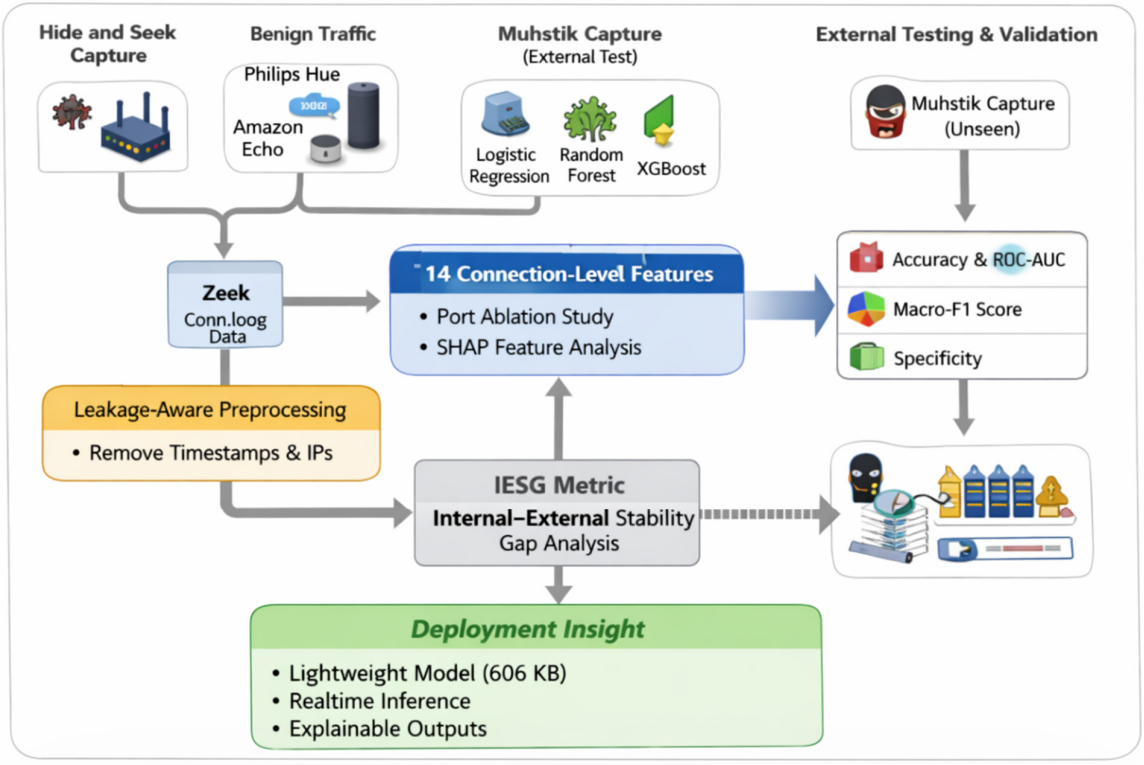


Figure 1. The proposed system workflow.

3.1. Data Selection

IoT-23 was selected as the empirical foundation because it provides labeled network traffic derived from actual IoT devices and concrete malware executions [4]. To preserve reproducibility while maintaining a clear methodological focus, four compact scenarios were selected rather than the full archive. The development pool consisted of three captures:

one malicious—CTU-IoT-Malware-Capture-1-1 (Hide and Seek)—and two benign—CTU-Honeypot-Capture-4-1 (Philips Hue bridge) and CTU-Honeypot-Capture-5-1 (Amazon Echo). Separately, a second malicious scenario—CTU-IoT-Malware-Capture-3-1 (Muhstik)—was reserved exclusively for external evaluation. This strict separation of scenarios ensures that the development pool and the external evaluation scenario remain distinct, establishing a clear boundary between model development and stress evaluation.

Table 2 summarizes the selected scenarios and their class distributions. The development set, which combined Hide and Seek and two benign scenarios, contained 8834 flows. The reserved Muhstik scenario, used only for external evaluation, comprised 156,103 flows. The external evaluation was intentionally demanding due to a marked class imbalance and behavioral divergence from the Hide and Seek malware used for training. Thus, strong internal performance suggests only preliminary learning effectiveness, not robust transferability.

Table 2. IoT-23 scenarios used in the research.

Scenario	Role	Flows	Benign	Malicious	Traffic Type/ Malware Family
CTU-IoT-Malware-Capture-1-1	Development	7008	3000	4008	Hide and Seek
CTU-Honeypot-Capture-4-1	Development	452	452	0	Philips Hue bridge
CTU-Honeypot-Capture-5-1	Development	1374	1374	0	Amazon Echo
CTU-IoT-Malware-Capture-3-1	External stress test	156,103	4536	151,567	Muhstik

In addition to the primary IoT-23 evaluation, supplementary held-out validation experiments were conducted using publicly accessible network trace datasets specifically containing DarkNexus and generic malware traffic. These supplementary datasets served only as robustness-oriented external validation and were deliberately excluded from training or developing the initial model. Their inclusion enabled a secondary assessment of how well behavioral patterns detected in Muhstik generalize to other contexts.

All network traffic captures analyzed in this study were sourced exclusively from the public IoT-23 repository published by Garcia et al. [4]. No proprietary or self-collected data was introduced at any stage of the analysis. The specific scenarios were intentionally selected to construct a compact yet behaviorally diverse evaluation environment, comprising one malicious development scenario, two benign development scenarios, and a single, fully held-out malicious scenario reserved exclusively for external stress testing. Consequently, scenario selection was driven by three primary criteria: (i) reproducibility, (ii) a manageable computational footprint, and (iii) a strict demarcation between the development and external evaluation data distributions.

Only the Zeek conn.log.labeled files associated with each scenario were utilized. No payload extraction, packet reconstruction, synthetic data augmentation, or relabeling operations were conducted. The labels in the original IoT-23 dataset were retained and directly mapped to a binary classification scheme (benign versus malicious), without altering the underlying attack semantics. Feature preprocessing was strictly limited to leakage-aware feature removal, the explicit handling of missing-value placeholders (e.g., '-' and '(empty)'), median imputation for numerical variables, mode imputation for categorical variables, and one-hot encoding of categorical attributes.

To ensure reproducibility, all randomized operations—including train/validation/test dataset splitting and benign-class downsampling—were executed using fixed random seeds. Stratified splitting was employed, allocating 60% of the data to training, 20% to validation, and 20% to internal testing. The external evaluation utilizing the Muhstik

scenario was conducted without any retraining, recalibration, or data mixing. Furthermore, no scenario-specific preprocessing steps were introduced beyond the unified pipeline detailed in Section 3.2.

3.2. Feature Policy and Preprocessing

Only the Zeek conn.log.labeled output files were utilized as input for this study. These files provided a connection-level perspective on the recorded network activity. After pruning non-contributing features, 14 variables were retained to characterize communication behavior and flow structure: src_port, dst_port, protocol, service, duration, orig_bytes, resp_bytes, state, miss_bytes, history, orig_pkts, orig_ip_bytes, resp_pkts, and resp_ip_bytes. Timestamps, raw IP addresses, flow IDs, tunnel_parents, and local_orig were explicitly excluded from consideration. Although these features do not serve as formal labels, they can directly or indirectly reveal the capture identity, thereby artificially inflating performance metrics. Consequently, feature inclusion and exclusion were treated as integral components of the experimental protocol, rather than mere data preprocessing steps.

Placeholder values, specifically '-' and '(empty)', were explicitly designated as missing values. The remaining missing numeric values were imputed using the median of their respective columns within the training fold. Similarly, missing categorical values were imputed using the mode of their respective columns within the training fold. Subsequently, columns containing categorical data were one-hot encoded. The combined development dataset was partitioned using stratified sampling, allocating 60% to training, 20% to validation, and 20% to internal testing. The raw training split comprised 2895 benign flows and 2405 malicious flows. To mitigate the potential adverse effects of learning from a highly imbalanced dataset, the benign class within the training portion was randomly downsampled to exactly match the 2405 malicious flows. The validation and internal test sets remained unmodified.

3.3. Model Set

Model selection was deliberately kept narrow to align with the study's principal focus. Linear Logistic Regression was established as the baseline benchmark due to its high interpretability and relatively low memory footprint. Random Forest was included to provide a bagged tree ensemble, offering enhanced capability for modeling non-linearity, albeit at the potential cost of increased memory usage. The included boosted-tree models, XGBoost and LightGBM, have historically demonstrated strong performance on structured intrusion detection datasets and are orders of magnitude more lightweight than deep neural networks [10,11,16,17].

Hyperparameter tuning was also kept simple. Bad settings were avoided, but the study did not test every possible option. All models were trained to classify traffic as normal or an attack. The validation set helped spot failures and make small adjustments. The models were first tested on a regular internal test set, then on a separate scenario called Muhstik. Finally, we tested on an external dataset, which is a tougher way to assess whether the models work in new situations than just internal rankings.

3.4. Evaluation Protocol

The evaluation protocol comprised both predictive and operational metrics. For predictive performance, the following metrics were selected: accuracy, precision, recall, F1 score, ROC-AUC, macro-F1, specificity, and balanced accuracy. Each was chosen to capture different aspects of model behavior; for example, ROC-AUC addresses trade-offs between true and false positives, and macro-F1 mitigates the effects of class imbalance, especially during cross-scenario evaluation. This diverse set prevents misleading conclusions that

could result from using accuracy alone. Additionally, operational metrics—including model size (measured by the number of parameters) and inference latency per 1000 flows—were recorded to reflect practical deployment considerations. These operational properties were included because an intrusion detector may not be suitable if it is too large, slow, or uninterpretable for gateway-scale deployment.

Figure 2 outlines the evaluation workflow. Model performance was first evaluated using the internal hold-out split of the development data. Trained models were then tested directly on the external Muhstik scenario with no retraining or recalibration. This phase served as a robustness test, showing whether a model's learned decision boundaries generalized to broader malicious patterns or simply overfitted to the development scenario.

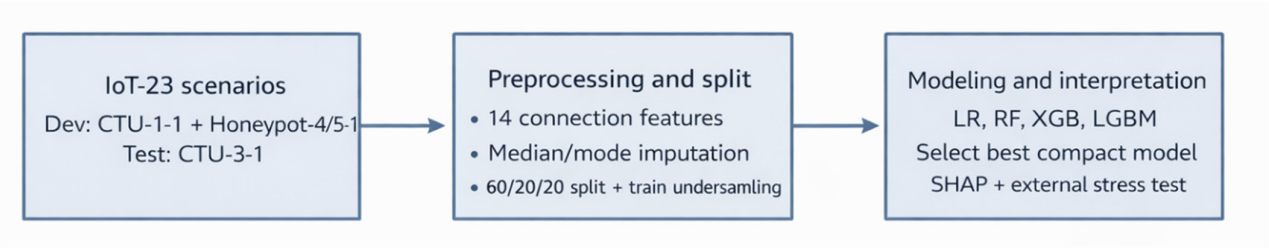


Figure 2. Compact IoT-23 modeling and external stress testing: Experimental workflow.

3.5. Explainability Stage

Following model selection, a SHAP (Shapley Additive exPlanations) analysis was conducted on the highest-performing compact model to evaluate global and local feature contributions [7,14]. To maintain interpretability following one-hot encoding, the mean absolute SHAP values were aggregated back into their original feature categories. This aggregation ensured that the resulting global feature importance analysis remained highly interpretable.

The objective of this step extended beyond the mere reporting of feature importance. SHAP was additionally used to determine whether the selected model relied on semantically relevant traffic features or had inadvertently learned unanticipated, narrow structural artifacts. If the model's performance had been heavily reliant on features associated with scenario-specific structures, confidence in its cross-scenario transferability would be significantly diminished. Therefore, the explainability results served as a critical verification step to ensure alignment between the model's predictive performance and expected network traffic behavior.

4. Results

This section presents the experimental findings and evaluates the overall efficacy of the proposed system.

4.1. Internal Hold-Out Performance

Table 3 summarizes the results from the internal hold-out split. As anticipated, given their performance on the pooled development data, all four models demonstrated highly robust predictive capabilities. Logistic regression achieved an F1-score of 0.9931, indicating a significant discriminatory signal inherent to the 14 selected connection-level features. Further performance improvements were observed when utilizing tree-based models. LightGBM, in particular, delivered the strongest performance on the internal split, achieving accuracy, F1-score, and ROC-AUC values of 0.9994, 0.9994, and 1.0000, respectively. While these metrics approach the theoretical ceiling for this task, they indicate that the

subtle structural patterns learned through non-linear modeling were indeed present and identifiable within the development scenarios.

Table 3. Internal holdout performance on the pooled development scenarios.

Model	Accuracy	ROC-AUC	Precision	Inference ms/1k	Recall	F1
LR	0.9938	0.9991	0.9925	4.3412	0.9938	0.9931
RF	0.9977	1.0000	0.9963	87.1529	0.9988	0.9975
XGB	0.9983	1.0000	0.9963	6.4737	1.0000	0.9981
LGBM	0.9994	1.0000	0.9988	7.8763	1.0000	0.9994

Nevertheless, such internal metrics must be interpreted with caution. While they confirm that the selected feature policy and modeling framework are exceptionally well-suited to the provided development captures, they do not inherently guarantee system robustness against distribution shifts. Consequently, evaluation on the fully held-out Muhstik scenario takes absolute precedence over these internal rankings for assessing true operational viability.

4.2. External Stress Test

An evaluation of the fully held-out Muhstik scenario significantly altered the practical ranking of the models. As detailed in Table 4, LightGBM demonstrated the most robust generalization, achieving the highest external accuracy (0.8341), the strongest ROC-AUC (0.8600), and the peak positive-class F1-score (0.9083). XGBoost followed with an external F1-score of 0.8925, whereas Logistic Regression and Random Forest converged to nearly identical F1-scores of 0.8856.

Table 4. External stress-test performance on CTU-IoT-Malware-Capture-3-1 (Muhstik).

Model	Accuracy	ROC-AUC	Precision	Inference ms/1k	Recall	F1	Specificity	Macro-F1	Balanced Acc.
LR	0.7976	0.8238	0.9813	2.3503	0.8069	0.8856	0.4863	0.5041	0.6466
RF	0.7977	0.8528	0.9814	11.9285	0.8069	0.8856	0.4879	0.5043	0.6474
XGB	0.8085	0.7827	0.9806	3.9647	0.8189	0.8925	0.4597	0.5075	0.6393
LGBM	0.8341	0.8600	0.9802	3.8939	0.8463	0.9083	0.4288	0.5195	0.6375

Although these findings indicate that boosted-tree methods maintained a relative advantage across scenario shifts, macro-F1 and specificity provided clearer insight into overall model performance by accounting for both class distribution and correct identification of negatives. Specifically, for LightGBM, the macro-F1 score declined sharply to 0.5195, and specificity dropped to 0.4288, despite achieving a comparatively strong positive-class F1-score. The macro-F1 emphasizes balance across classes, while specificity indicates the model's reliability in correctly identifying benign flows. The confusion matrix presented in Figure 3 further illustrates this pronounced imbalance. While the model successfully identified a substantial proportion of malicious flows, it simultaneously generated 2591 false positives and failed to detect 23,301 malicious instances. Consequently, the external evaluation reveals a state of partial transferability: although the learned decision boundary successfully captured meaningful malicious structural patterns, its cross-class calibration degraded significantly when confronted with a novel behavioral profile.

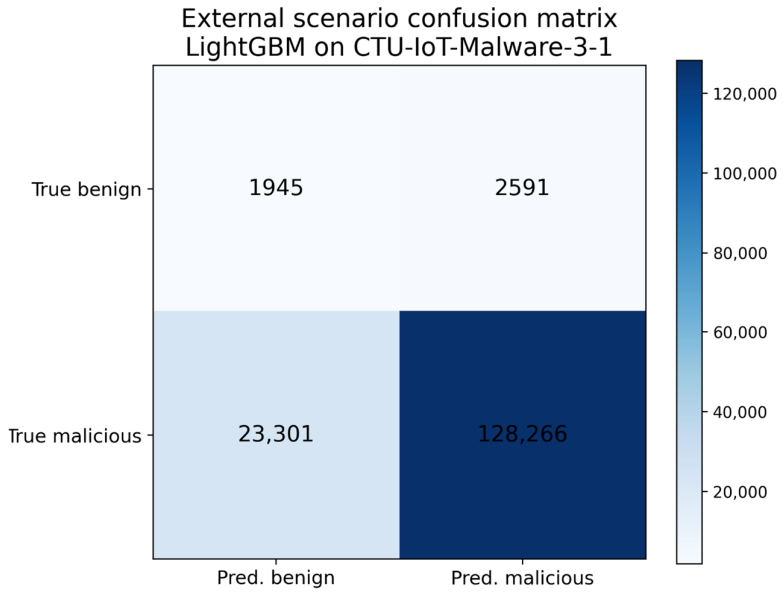


Figure 3. Confusion matrix for the selected LightGBM model evaluated on the fully held-out Muhstik scenario.

4.3. Footprint and Inference Trade-Off

Figure 4 plots internal F1 performance against serialized model size. This provides a concise view of the predictive–operational trade-off. The comparison is especially pertinent for gateway-oriented deployments, where strict memory and latency constraints may outweigh minor gains in predictive accuracy. Logistic Regression emerged as the most compact model, with a serialized size of merely 4.4 KB, and exhibited the lowest inference latency. However, it simultaneously yielded the weakest predictive performance among the evaluated models. While the Random Forest algorithm improved internal accuracy relative to the linear baseline, this improvement came at a substantial cost in memory footprint. With a serialized size of approximately 1.49 MB, it was the largest model evaluated; furthermore, its inference cost was noticeably higher than that of the boosted-tree alternatives. XGBoost occupied an intermediate position. At a serialized size of 256.9 KB, it remained highly compact while achieving a robust internal F1-score of 0.9981. Although LightGBM was larger at 606.3 KB, it remained sufficiently lightweight for practical gateway-side deployment and delivered the strongest predictive performance. LightGBM was selected over XGBoost because its superior predictive quality and acceptable model size offered the optimal tradeoff among detection quality, memory footprint, and inference efficiency.

4.4. Explainability Results

Figure 5 illustrates the SHAP profile of the selected LightGBM model. The destination port had the greatest impact on model predictions. Connection history and protocol followed in significance, while the source port ranked fourth. Conversely, packet- and byte-based variables exerted a comparatively minor influence on the decision boundary.

This observed pattern is highly consistent with the characteristics of compact IoT malware scenarios. Malicious traffic frequently deviates from benign baselines in its port utilization, transport-layer behavior, and connection-state transitions. Zeek's history field effectively captures part of this stateful behavior. The high importance assigned to these variables closely matches expected network dynamics.

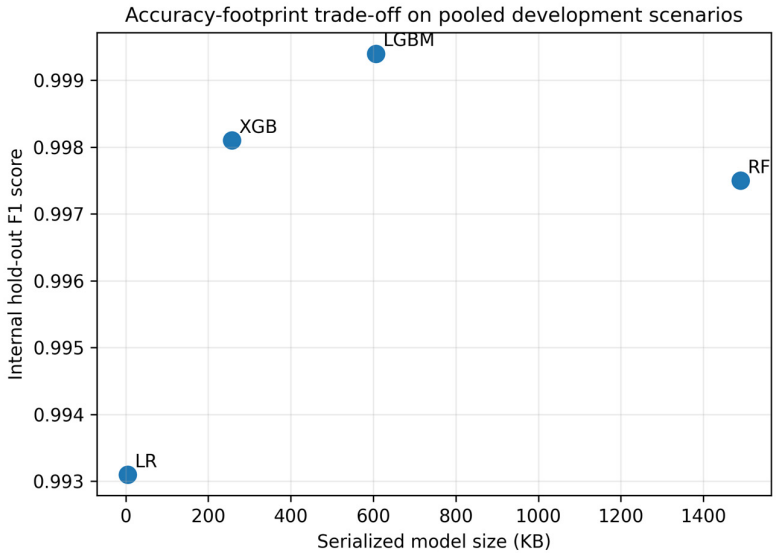


Figure 4. Internal F1 versus serialized model size for the four candidate models.

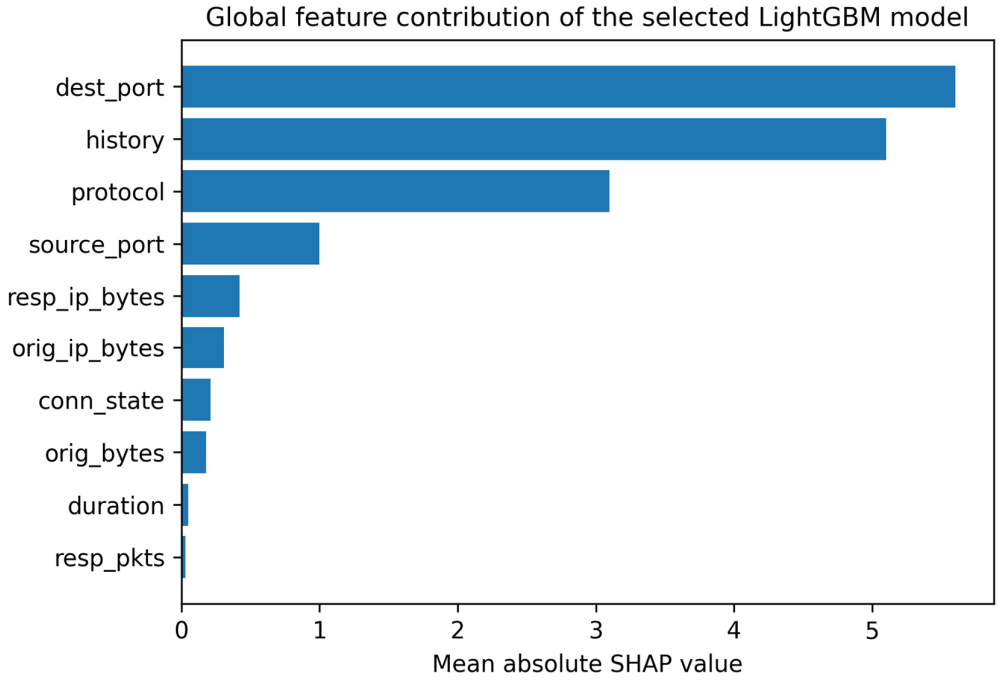


Figure 5. Global mean absolute SHAP values for the selected LightGBM model.

Simultaneously, these findings warrant caution. While port-related variables help elucidate why the model retains utility within the external Muhstik scenario—indicating that certain fundamental communication patterns remain informative despite scenario shifts—a portion of the model's predictive capacity may still rely on scenario-specific communication artifacts rather than universally generalized malicious behavior. For this reason, the near-perfect internal performance metrics are interpreted with appropriate caution.

4.5. Feature Robustness and Port-Ablation Analysis

The destination and source ports were among the features with the highest SHAP values in the selected LightGBM model. The prominence of these predictors aligns with standard intrusion-detection intuition, as malicious traffic is known to generate anomalous flows regarding the specific ports and services utilized. However, their strong predictive capacity also raises a concern about transferability: port activity can depend heavily

on user behavior, malware variants, and specific device configurations. Consequently, models that rely heavily on port-level features may generalize less effectively to novel operational distributions.

A simple feature-ablation experiment was conducted to better understand this phenomenon. Specifically, a LightGBM model was trained after completely removing the source and destination port variables from the feature set. No other components of the analysis workflow were altered; data preprocessing, dataset splitting, and hyperparameter selection remained identical. Furthermore, no supplementary tuning or structural modifications were applied to the model. Consequently, any observed differences in the model's performance can be directly attributed to its reduced reliance on port-level information.

Table 4 presents the internal and external evaluations of this ablated model, alongside the results of the original LightGBM workflow previously detailed in Table 3. The deletion of these two features reduced the internal F1-score by 0.0122, indicating that port information indeed facilitates internal discrimination within the development dataset. The external weighted F1-score decreased slightly; however, the macro-F1 and specificity metrics remained unchanged or improved.

Table 5 summarizes the internal and external performance comparisons between the original LightGBM model and its port-ablated variant. Removing the two port-level variables reduced the internal F1-score from 0.9994 to 0.9872. This confirms that port information contributes substantially to class discrimination in development scenarios. The external positive-class F1-score also declined, but only moderately. In contrast, both the macro-F1 score and specificity improved with this reduced feature configuration.

Table 5. Port-ablation analysis for LightGBM (internal and external evaluation).

Model Variant	Internal F1	External Macro-F1	External Specificity	External F1
Full feature set	0.9994	0.5195	0.4288	0.9083
Without ports	0.9872	0.5413	0.5036	0.8721

This behavior thus does not reflect an across-the-board performance decline, but rather indicates a reshuffling of predictive focus between the classes. In particular, it is hypothesized that under the original decision boundary, connections were more likely to be classified as malicious if their port utilization mimicked patterns observed in the development captures. Following the reduction in port predictor influence, connection type, past connection history, and traffic-volume metrics (such as packet and byte counts) became proportionally more influential. This revised decision boundary was less skewed toward the malicious class, thereby improving the identification of benign flows within the external Muhstik dataset.

Taken together, the ablation analysis indicates not a simplistic causal relationship, but rather that port-level attributes serve as highly informative features that help maintain elevated malicious-flow detection accuracy even outside the initial training scenario. Simultaneously, the model's reliance on scenario-dependent communication patterns adversely affected its specificity when confronted with a distribution shift. It can therefore be concluded that the external stability deficit is not driven by a single feature shortcut; nevertheless, port-level information significantly factors into the model's internal–external performance trade-off. This trade-off is clearly illustrated in Figure 6: whereas the original model achieved superior internal and external positive-class F1 Scores, the port-ablated variant sacrificed these metrics in favor of higher macro-F1 and specificity.

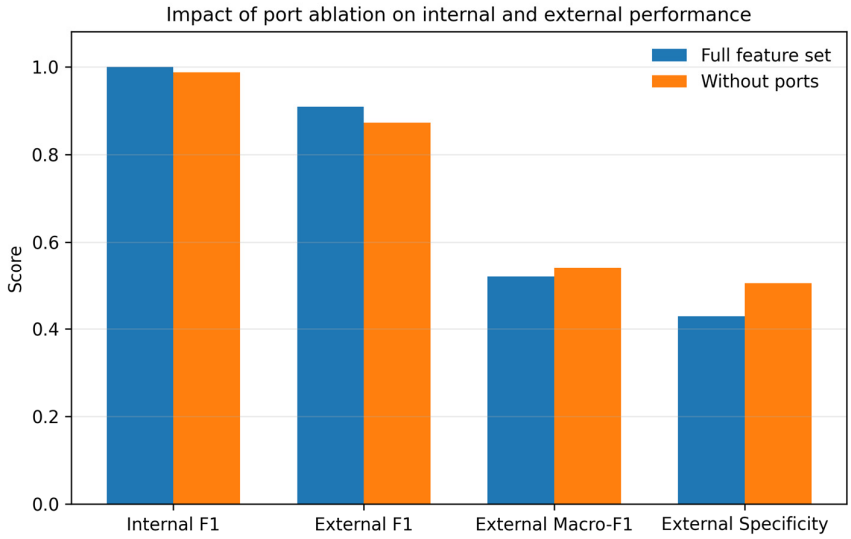


Figure 6. When port variables are removed from the full LightGBM model, there is a slight decrease in internal and external positive-class F1 scores, but a corresponding improvement in external macro-F1 and specificity.

4.6. Internal–External Stability Gap Performance Analysis

Standard assessment metrics such as accuracy, F1-score, and ROC-AUC are valuable for evaluating predictive performance. However, they do not directly measure model robustness under distribution shifts. This limitation is critical in IoT intrusion detection, as traffic can vary across devices, locations, and malware variants. Thus, strong in-sample performance may not ensure stable generalization in real-world environments.

The evaluation methodology is enhanced to address this limitation by incorporating a stability-oriented analysis that explicitly quantifies performance degradation between internal validation and external testing settings. This paper introduces the Internal–External Stability Gap (IESG), formalizes it, and extends it to better reflect generalization behavior. The original IESG is defined as the absolute difference between the internal and external F1-scores for the positive (malicious) class. This formulation, while intuitive, fails to account for the relative magnitude of the performance decline or the broader implications of class imbalance. Therefore, a normalized version of this metric is presented as follows:

$$\boldsymbol{IESG} = |\boldsymbol{F1_internal} - \boldsymbol{F1_external}|$$

where $F1_{internal}$ is the positive-class F1-score obtained on the internal hold-out test set, and $F1_{external}$ is the positive-class F1-score obtained on the external unseen scenario.

Normalized IESG:

$$IESG_{norm} = \frac{|\ F1_{internal} - F1_{external}\ |}{F1_{internal} + \epsilon} \quad (1)$$

where $\epsilon = 10^{-6}$ is a small constant introduced to avoid division instability.

This normalized formulation makes it easier to compare models with varied baseline performance levels in a more uniform way. Lower values mean that generalization is stronger and the capacity to detect changes in distribution is better. We also broaden the stability study by adding a multi-dimensional stability view to it. We specifically look at changes in macro-F1 and specificity coupled with the F1 score for the positive class. This lets us measure both the stability of malicious detection performance and the balance between benign and malicious classification when things change.

The Multi-Metric Formulation:

$$IESG_{multi} = \alpha \Delta F1 + \beta \Delta MacroF1 + \gamma \Delta Specificity$$

where

$$\Delta F1 = \mid F1_{internal} - F1_{external} \mid$$

$$\Delta MacroF1 = \mid MacroF1_{internal} - MacroF1_{external} \mid$$

$$\Delta Specificity = \mid Specificity_{internal} - Specificity_{external} \mid$$

and:

$$\alpha + \beta + \gamma = 1$$

In this study:

$$\alpha = \beta = \gamma = \frac{1}{3}$$

Unlike standard robustness evaluation, which relies only on external accuracy or F1-score, the proposed framework measures how model performance changes between internal and external conditions. This approach captures both predictive quality and stability under distribution shifts. Including macro-F1 and specificity also helps avoid bias toward the dominant malicious class. As a result, the metric offers a clearer view of model reliability for IoT intrusion detection systems as traffic conditions evolve.

The proposed IESG framework should not be interpreted as an attempt to replace established concepts such as out-of-distribution generalization analysis, cross-domain validation, calibration error, or classical generalization-gap assessment. Instead, its purpose is more operationally focused. Traditional generalization gaps typically compare training and testing performance within a shared dataset distribution, whereas the proposed framework explicitly quantifies degradation between internal validation conditions and fully separated external traffic scenarios. Similarly, calibration-oriented metrics evaluate probability reliability, while IESG focuses directly on predictive retention stability under a deployment-oriented scenario shift.

The primary contribution of the proposed formulation, therefore, lies not in introducing a fundamentally new theoretical learning metric but in providing a compact and deployment-oriented stability interpretation tailored to IoT intrusion detection evaluation. In particular, the multi-metric extension integrates positive-class retention, benign-flow discrimination, and class-balanced degradation into a unified stability-oriented analysis framework. This enables clearer diagnosis of situations where high malicious-detection performance coexists with deteriorating specificity or macro-level balance under external operating conditions. Consequently, the framework is intended as a complementary robustness-analysis tool rather than as a replacement for broader distribution-shift or generalization theory.

This extended formulation provides three principal advantages. First, it distinguishes between models that maintain high malicious-flow detection at the expense of benign classification and those that achieve more balanced performance. Second, it facilitates a more nuanced interpretation of robustness beyond single-metric evaluations. Third, it aligns the evaluation process with practical deployment considerations, where both detection accuracy and strict false-positive control are critical.

All F1-scores are reported specifically for the positive (malicious) class, as the primary focus is to maintain consistency during external evaluation and to emphasize the stability analysis regarding malicious-flow detection retention. A lower IESG value indicates that detection quality is better preserved across diverse scenarios. This suggests that the learned decision boundary has successfully abstracted behavioral characteristics that generalize

beyond the development data distribution. Conversely, a larger gap indicates that detection performance is heavily dependent on the specific properties of individual scenarios, leading to reduced stability across varying deployment contexts. The IESG should not be considered a replacement for metrics such as macro-F1 or balanced accuracy. Instead, its objective is to provide supplementary insight by explicitly isolating the performance retention of the malicious class.

This stability-oriented evaluation framework is also integrated into the experimental pipeline as a complementary assessment to standard predictive metrics. It does not replace traditional evaluation measures; instead, it adds a diagnostic layer that highlights the sensitivity of learned decision boundaries to scenario variations. This consideration is particularly relevant for lightweight IoT intrusion detection systems, where compact models may be more susceptible to learning scenario-specific patterns.

4.7. Analysis of Extended Stability

The stability-oriented evaluation framework detailed in Section 4.6 was utilized to further investigate model robustness under distribution shifts. Alongside the original Internal–External Stability Gap (IESG), the normalized stability gap and the multi-metric stability indicator were also considered, offering a more thorough assessment. These measurements collectively show that predictive performance degrades when moving from internal to external validation, as summarized in Table 6. Notably, while the original IESG values suggest only minor differences among models, the normalized and multi-metric measures uncover more nuanced behavioral differences.

Table 6. Extended Stability Metrics across models.

Model	IESG	$IESG_{norm}$	ΔMacro-F1	ΔSpecificity	$IESG_{multi}$
LR	0.1075	0.1082	0.4859	0.5037	1.0971
RF	0.1119	0.1122	0.4857	0.5021	1.0997
XGB	0.1056	0.1058	0.4825	0.5303	1.1184
LGBM	0.0911	0.0912	0.4705	0.5612	1.1228

The normalized stability gap demonstrates that even models achieving near-perfect internal performance experience a significant decline when subjected to scenario shifts. LightGBM achieved the highest internal F1-score and the lowest normalized stability gap, indicating superior generalization capabilities relative to the alternative models. However, an examination of the multi-metric stability indicator reveals more pronounced performance disparities among the models. The results indicate that models with comparable positive-class F1-score retention may exhibit divergent behaviors when evaluated across multiple metrics. Although LightGBM achieved the highest external F1-score, it also experienced substantial declines in specificity and macro-F1, indicating an imbalance in its classification behavior. The composite stability metric captures this phenomenon more effectively than the original IESG.

Conversely, models such as Logistic Regression and Random Forests demonstrated slightly greater stability in specificity, despite their lower overall detection efficacy. This implies that simpler models may establish more conservative decision boundaries that generalize more uniformly across classes, albeit at the expense of diminished sensitivity to malicious traffic. These findings underscore the importance of employing a multi-dimensional approach when evaluating IoT intrusion detection technologies. Relying solely on positive-class F1 Scores or accuracy metrics can yield overly optimistic assessments of deployment readiness. The expanded stability framework demonstrates that high

detection rates can co-occur with significant degradation in benign-class identification under distribution shifts.

Furthermore, the findings offer deeper insights into the significance of feature dependence for model generalization. As evidenced by the port-ablation study, models that rely heavily on high-impact variables, such as port information, tend to perform well on internal and malicious-class metrics but exhibit reduced stability in specificity. Evaluating this trade-off using the proposed multi-metric stability indicator clarifies this dynamic.

Table 6 presents the Internal–External Stability Gap metrics for the four evaluated models. LightGBM exhibited the strongest retention, achieving the lowest gap value of 0.0911. Notably, LightGBM also achieved the highest internal and external F1-scores for the positive class. Compared to alternative models, this boosted-tree approach proved more adept at learning decision rules that generalize to contexts beyond the initial Hide-and-Seek development scenario.

Overall, this extended analysis demonstrates that model stability is a multifaceted construct, encompassing performance retention, class equilibrium, and sensitivity to feature distributions. The proposed evaluation framework facilitates a more comprehensive and interpretable assessment of these dynamics, thereby aiding in the selection of optimal models for real-world IoT deployment scenarios.

4.8. Supplementary Held-Out Malware-Family Validation

A supplementary held-out malware-family validation experiment was conducted using an additional network-trace dataset to further investigate the robustness of the proposed framework beyond the initial external Muhstik evaluation. The primary objective of this experiment was to determine whether the learned decision boundaries remained stable when evaluated against malware families not present during the original model development phase. Additionally, an expanded benign-traffic configuration was introduced to determine whether increasing the diversity of the benign class could enhance specificity and mitigate false-positive sensitivity under external testing conditions.

Table 7 summarizes the findings from this extended evaluation. The model performed exceptionally well on the held-out DarkNexus scenario. It achieved near-perfect scores across all metrics, including accuracy (the proportion of correct predictions), F1-score (the harmonic mean of precision and recall), macro-F1 (average F1-score across all classes), specificity (the proportion of true negatives correctly identified), and balanced accuracy (average of sensitivity and specificity). These results show that the learned feature representation effectively captured behavioral characteristics that generalize to this distinct malware family.

Table 7. Additional generalization validation results.

External Test Dataset	Accuracy	ROC-AUC	Precision	Recall	F1	Macro-F1	Specificity	Balanced Acc.
DarkNexus + benign	0.9998	1.0000	0.9998	0.9995	0.9996	0.9997	0.9999	0.9997
Generic + benign	0.9779	0.2400	0.9000	0.0384	0.0736	0.5312	0.9999	0.5191
DarkNexus + benign	0.9998	1.0000	0.9999	0.9995	0.9997	0.9998	1.0000	0.9997
Generic + benign	0.9775	0.1080	0.9091	0.0213	0.0417	0.5151	1.0000	0.5106

Evaluation of the Generic malware scenario revealed much weaker transfer performance. Specificity stayed high, but both recall and the positive-class F1-score dropped sharply. This disparity shows the model preserved benign-flow discrimination but missed many malicious flows from the Generic family. Thus, the decision boundary still depended partly on training malware traffic characteristics.

Integrating expanded benign traffic slightly improved benign-flow preservation and specificity, but did not significantly improve malicious-flow recall in the Generic scenario. This observation supports the study's broader conclusion: lightweight intrusion detection models can perform robustly under some distribution shifts but remain highly sensitive to new behavioral variations in novel malware families.

Table 8 highlights the stability of lightweight intrusion detection models under various external validation conditions. Lower values indicate less degradation in internal and external Macro-F1 performance, indicating stronger robustness to distribution shifts. LightGBM consistently has the lowest stability gap across Muhstik, DarkNexus, and Generic malware families. Its average of 0.2526 demonstrates the strongest overall balance in classification performance under unseen conditions. XGBoost shows the next-best stability, while Logistic Regression has the largest average degradation and is more sensitive to changes in the external distribution. Stability gaps under DarkNexus and Generic malware-family evaluations are generally smaller than those under the Muhstik scenario-shift, indicating that network scenario changes may create stronger distribution mismatches than some malware-family transitions. These findings emphasize the need to evaluate IoT intrusion detection systems across multiple external variations rather than relying solely on internal hold-out validation.

Table 8. Macro-F1 Stability ($IESG_{F_1}$) Across Malware-Family Validations.

Model	Primary External Test (Muhstik Scenario Shift)	External Test 1 (DarkNexus Dataset)	External Test 2 (Generic Dataset)	Average $IESG_{F_1}$ (All External Tests)
Logistic Regression	0.3173	0.2966	0.2915	0.3018
Random Forest	0.2889	0.2823	0.2823	0.2845
XGBoost	0.2695	0.2617	0.2575	0.2629
LightGBM	0.2648	0.2500	0.2431	0.2526

Figure 7 shows how the Macro-F1 scores of the evaluated models change across internal validation, the main Muhstik scenario-shift test, and the extra malware-family tests using the DarkNexus and Generic datasets. All models scored nearly perfect Macro-F1 on internal validation, meaning they learned well from the training data. However, their performance dropped across all external tests, indicating they are sensitive to changes in data and new traffic types.

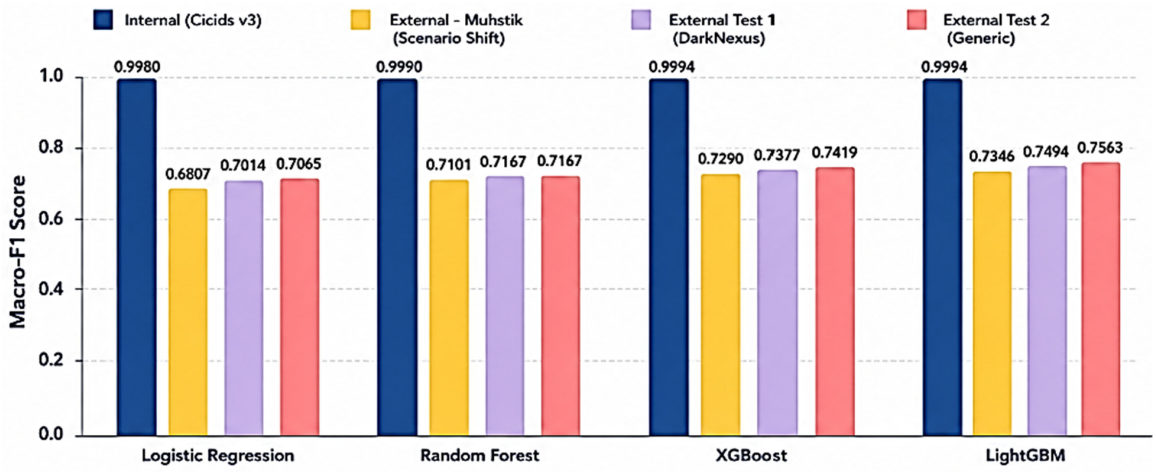


Figure 7. Macro-F1 performance comparison across internal evaluation, primary external scenario-shift testing (Muhstik), and supplementary held-out malware-family validations using the DarkNexus and Generic datasets.

The largest performance drop occurred in the Muhstik scenario-shift condition, where Macro-F1 values for the models ranged from 0.6807 to 0.7346. In comparison, across both DarkNexus and Generic malware-family validations, all models achieved slightly higher Macro-F1 scores. This outcome suggests that the effect of malware-family variation on Macro-F1 is less severe than the effect of broader scenario-level behavioral changes. Among the evaluated approaches, LightGBM consistently achieved the highest external Macro-F1 scores—0.7346 in Muhstik, 0.7494 in DarkNexus, and 0.7563 in Generic—outperforming the other models in these conditions. XGBoost ranked second in robustness, while Logistic Regression exhibited the greatest performance degradation under external evaluation. These results support the central finding of this study: lightweight boosted-tree models, especially LightGBM and, to a lesser extent, XGBoost, exhibit more stable performance than Logistic Regression in the presence of distribution shifts, although all models degrade under significant distribution shifts.

This supplementary validation experiment underscores the crucial need for a comprehensive evaluation of IoT intrusion detection systems across varied real-world conditions. While the proposed framework shows robustness against some held-out malware families, rigorous cross-dataset validation across heterogeneous datasets such as CICIoT2023 and ToN_IoT is essential before making claims about universal generalization.

5. Performance Analysis

The evaluated compact models achieved near-perfect internal accuracy, but this did not guarantee stable cross-scenario performance. All four models reached an F1-score of 0.99 on the internal holdout split, validating the hypothesis that the 14 selected connection-level features capture meaningful class-separating information within the development distribution. However, during the external Muhstik evaluation, it became clear that strong internal results did not ensure balanced performance in other scenarios. These findings demonstrate that a high positive-class F1-score can coexist with weaker benign accuracy and reveal non-negligible stability gaps when testing moves beyond the original development distribution.

LightGBM exhibited the optimal overall trade-off. It achieved both the highest external positive-class F1 Score and the highest ROC AUC score. Furthermore, it demonstrated the lowest Internal–External Stability Gap (IESG), suggesting superior transfer retention capabilities relative to the alternative models. Simultaneously, its external macro-F1 score and specificity remained only moderate. This reiterates that a high malicious-flow detection rate does not guarantee balanced benign-versus-malicious discrimination under an operating-context shift.

The port-ablation experiment offers partial context for this outcome. The results show that port variables were essential for internal discrimination and helped support external malicious detection. However, including these variables ultimately reduced both specificity and macro-F1 during distribution shifts. This does not mean port features are inherently harmful. Instead, their strong predictive power may tie the model more closely to the specific patterns of the development dataset.

The stability-gap analysis provides an additional perspective for interpreting these results. Although all evaluated models achieved comparable internal F1-scores, significant variance was observed in the extent of performance degradation during cross-scenario evaluation. This indicates that evaluating and comparing classifiers relying solely on a single holdout split is insufficient for making realistic deployment decisions. Therefore, stability is treated as an orthogonal metric for characterizing performance, as it effectively captures the degree to which learned decision boundaries align with the initial development distribution. These concerns are echoed by emerging literature advocating robust, realistic

IoT IDS evaluations that account for practical deployment, particularly in resource-efficient and federated contexts [8,12,13,19,20].

Finally, these findings have direct operational implications. The evaluated detector is not designed for production-ready autonomous analysis. It is meant to be a lightweight triage tool that can operate under strict resource limits. During the Muhstik evaluation scenario, the chosen model produced 2591 false positives and missed 23,301 malicious flows.

Despite strong results in detecting harmful network flows, the model's observed external specificity of 0.4288 indicates it generates many false alerts when faced with changes in real-world conditions. Whether this trade-off is acceptable in real IoT systems depends on the specific goal. Security-focused monitoring tools usually prioritize finding as many attacks as possible to avoid missing any, since missed threats pose a greater risk than harmless alerts. In these cases, having lower specificity may be fine for early warning or sorting a network. On the other hand, in busy IoT setups, too many false alerts can create extra work for analysts, erode their trust in the alerts, and limit the system's growth. can grow. Therefore, the results suggest the framework works best when set up to catch as many threats as possible, rather than for use without human review. Making this system work in real-world settings will likely need adjusting alert thresholds, adding more examples of safe traffic to the training process, using retraining methods that respond to changes, or adding extra steps to filter out harmless flows. These steps are needed to help the model tell safe from harmful flows without losing the ability to spot true threats.

This rate of incorrect benign classifications implies that manual review, threshold calibration, benign-set diversification, or model retraining would be necessary prior to production deployment. For edge-oriented IoT applications in particular, such a model is more likely to serve effectively in isolating suspicious flows rather than definitively certifying connections as safe or benign. Taking all results into consideration, this study effectively rebuts the notion that lightweight machine-learning models must inevitably sacrifice predictive capabilities in exchange for operational efficiency. Within the development distribution, a minimal trade-off was observed, as the 14 hand-crafted features were sufficient for even Logistic Regression to achieve an F1-score exceeding 0.99. A distinct trade-off was observed when introducing a scenario shift; however, this manifested as a balance between predictive performance and a combination of memory footprint, interpretability, and stability. Based on these metrics, LightGBM exhibited the optimal trade-off among the evaluated models. Nevertheless, its subpar stability further demonstrates that a lightweight architecture alone does not guarantee robustness or transferability. Previous research on lightweight ensemble models, semantically enriched features, and dimensionality reduction for IDSs similarly suggests that satisfying deployment demands requires actively balancing efficiency and transferability, rather than focusing on either property in isolation [21–25].

Comparative Analysis with Deep Learning-Based IDS Approaches

Further contextualization of the proposed framework is provided by comparing its characteristics with representative deep learning and hybrid intrusion detection approaches documented in recent IoT IDS literature [21–23]. Deep learning architectures—such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and hybrid models—are capable of automatically learning complex traffic representations and frequently achieve strong predictive performance on benchmark datasets. However, compared with lightweight machine learning models, these approaches typically introduce substantially higher computational complexity, larger memory requirements, longer training times, and reduced interpretability.

In contrast, the proposed framework was intentionally designed to prioritize lightweight deployment, high interpretability, and stability-oriented evaluation. The se-

lected LightGBM model achieved strong predictive performance while maintaining a compact memory footprint of approximately 606 KB and a low inference latency suitable for gateway-oriented IoT environments. Furthermore, integrating SHAP-based explainability enabled direct interpretation of feature influence, a process that remains inherently difficult in many opaque deep neural network architectures.

An additional critical distinction concerns the evaluation of robustness under distribution shifts. Many deep learning-based IDS studies primarily emphasize improvements in internal accuracy, whereas relatively few explicitly analyze stability degradation across unseen traffic scenarios. The proposed framework complements the existing literature by concurrently evaluating predictive performance and stability retention under external scenario variations.

Overall, the comparison detailed in Table 9 suggests that increasing architectural complexity does not inherently guarantee superior deployment-oriented robustness. Instead, lightweight boosted-tree models may provide a more practical balance among predictive performance, computational efficiency, explainability, and operational stability for resource-constrained IoT intrusion detection systems.

Table 9. Conceptual comparison between the proposed lightweight framework and representative deep learning-based IoT IDS approaches.

Property	Proposed Framework (LightGBM-Based)	Deep Learning/Hybrid IDS Approaches [21–23]
Model Type	Lightweight boosted-tree ensemble	CNN, LSTM, hybrid deep architectures
Feature Representation	Hand-crafted connection-level features	Automatically learned representations
Interpretability	High (SHAP-supported)	Limited/often post hoc
Training Complexity	Low	High
Inference Cost	Low	Moderate–High
Model Footprint	Compact (~606 KB)	Typically larger
Deployment Suitability	Gateway-friendly	Resource-intensive
Stability Evaluation	Explicit cross-scenario analysis	Rarely emphasized
Explainability Support	Native feature-level analysis	Often difficult to interpret

Figure 8 illustrates the practical trade-offs between lightweight and deep learning-based intrusion detection systems. Although deep neural architectures may achieve strong predictive performance, their increased computational and operational complexity can significantly limit deployment feasibility in resource-constrained IoT environments. In contrast, the proposed framework emphasizes a more balanced design paradigm, integrating robust detection capabilities with a compact memory footprint, high interpretability, and a stability-oriented evaluation methodology.

These conclusions must necessarily be qualified by the specific limitations of this study. An intentionally small development dataset was utilized, and the external evaluation captures were drawn from the same IoT-23 benchmark as the training data. Consequently, the stability gaps demonstrated were induced by controlled scenario shifts rather than broad dataset heterogeneity. The utilization of a more diverse benign pool, the incorporation of additional malware families, and the evaluation of cross-dataset generalizability using benchmarks such as CICIoT2023 or ToN_IoT would likely reveal further performance variability. With these considerations in mind, the current findings are presented as evidence that LightGBM achieves reasonable transferability under measurable scenario shifts; they do not, however, assert universal model generalization.

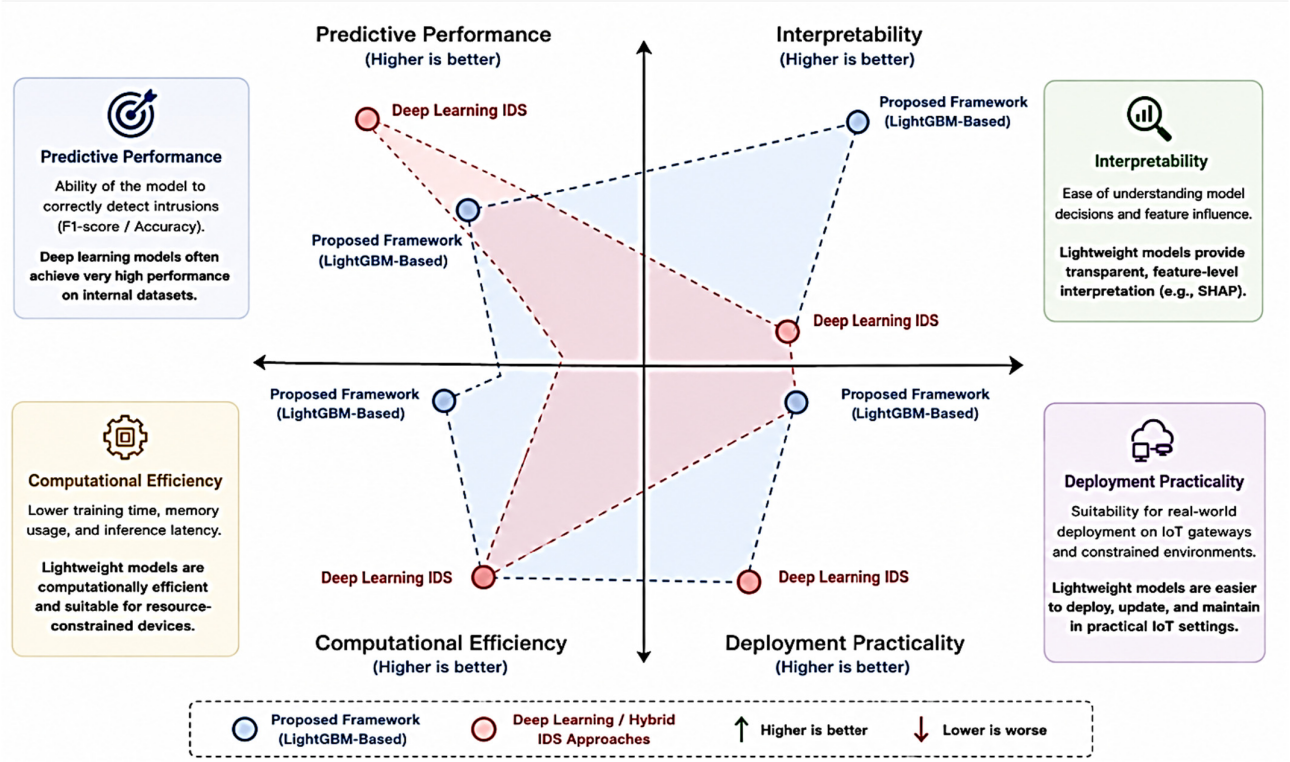


Figure 8. Conceptual comparison between lightweight boosted-tree intrusion detection models and representative deep learning-based IDS approaches in terms of predictive performance, interpretability, computational efficiency, and deployment suitability.

6. Conclusions

This paper introduces a lightweight, highly interpretable approach for detecting intrusions within IoT traffic, leveraging compact scenarios derived from the IoT-23 dataset. Four established machine learning models were trained on a deliberately restricted set of 14 connection-level features, rigorously omitting any identifiers susceptible to data leakage. During internal evaluation, all models demonstrated strong predictive performance, with LightGBM achieving the highest F1 Score on the holdout split. When subjected to the more rigorous external Muhstik stress-test scenario, LightGBM maintained its superiority, achieving an accuracy of 0.8341 and a positive-class F1-score of 0.9083, all while retaining a highly compact memory footprint of approximately 606 KB.

However, the primary implications of this study extend beyond these raw predictive metrics. The observed degradation in macro-F1 and specificity during the external evaluation demonstrates that high efficacy in detecting malicious traffic within a known scenario does not inherently guarantee reliable discrimination between benign and malicious behavior under novel operational conditions.

Overall, these results indicate that compact classical and boosted tree models are highly viable solutions for resource-constrained IoT environments. Nevertheless, practical deployment would require integrating more diverse benign traffic profiles, evaluating across a broader range of operational scenarios, and rigorously calibrating thresholds to mitigate performance degradation under distribution shifts. By explicitly decoupling internal performance metrics from cross-scenario generalization—and by formally quantifying the stability gap between them—this study underscores the critical need to evaluate intrusion detection models under conditions that accurately reflect realistic operational constraints.

Future research directions should incorporate expanded benign datasets, validate models on heterogeneous benchmarks such as CICIoT2023 or ToN_IoT, and further investigate threshold calibration techniques tailored to heavily imbalanced class distributions. Furthermore, assessing the resilience of these compact models against adversarial perturbations and data poisoning attacks, as well as evaluating their efficacy within federated learning architectures, remains an essential priority for future exploration [3,5,19–23]. Finally, supplementary validation on fully held-out malware families demonstrated that model robustness can vary substantially across unseen traffic behaviors, further reinforcing the imperative for deployment-oriented external evaluations that extend well beyond standard internal benchmark performance.

7. Practical Implications and Limitations

From an operational perspective, the results indicate that effective IoT intrusion screening does not inherently require deep or computationally intensive architectures. A carefully constrained connection-level representation, paired with a compact boosted-tree model, can deliver robust first-stage detection at modest memory and latency costs. The reported footprint and inference measurements suggest that LightGBM and XGBoost remain highly viable candidates for gateway-side deployment when predictive quality, inference speed, and interpretability must be balanced simultaneously.

Although serialized model size and inference latency provide useful indicators of computational efficiency, these metrics alone do not fully demonstrate deployment suitability for actual IoT gateways or embedded edge platforms. Because the current evaluation was conducted within a conventional experimental computing environment—rather than on physically constrained hardware such as Raspberry Pi devices, OpenWrt-based routers, or industrial embedded systems—the reported lightweight characteristics should be interpreted as relative efficiency indicators rather than definitive proof of edge-device deployability. Future research should therefore incorporate hardware-level benchmarking under realistic deployment conditions, encompassing empirical measurements of memory utilization, CPU load, thermal behavior, power consumption, and sustained real-time throughput on representative embedded platforms. Such an evaluation would provide a more rigorous validation of operational suitability for practical IoT gateway deployments.

Certain limitations of this study must be explicitly acknowledged. Specifically, the evaluation employs a compact subset of the IoT-23 dataset rather than the complete archive, formulates the intrusion detection task as binary classification rather than multi-class attack categorization, and evaluates transferability utilizing a single held-out scenario. Furthermore, the external testing environment remains confined to the same dataset family, and the adopted feature-extraction policy explicitly excludes payload-level and temporal-sequence modeling. Accordingly, these findings should not be broadly generalized to all IoT deployments or novel malware families without further empirical validation.

Although supplementary validation against held-out malware families was conducted, these experiments do not constitute a fully heterogeneous cross-dataset validation. This limitation arises because all evaluated captures were derived from related traffic-generation environments rather than entirely independent benchmark ecosystems.

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Data Availability Statement: The primary network traffic scenarios utilized in this study are publicly accessible via the IoT-23 repository. Supplementary held-out validation experiments targeting specific malware families were conducted utilizing publicly available trace datasets that contain DarkNexus and Generic malware traffic captures. Furthermore, the feature selection policy, preprocessing workflow, dataset partitioning strategy, model configurations, and stability analysis procedures have been detailed sufficiently to facilitate independent reproducibility with minimal ambiguity.

Conflicts of Interest: The authors declare no conflicts of interest.

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